

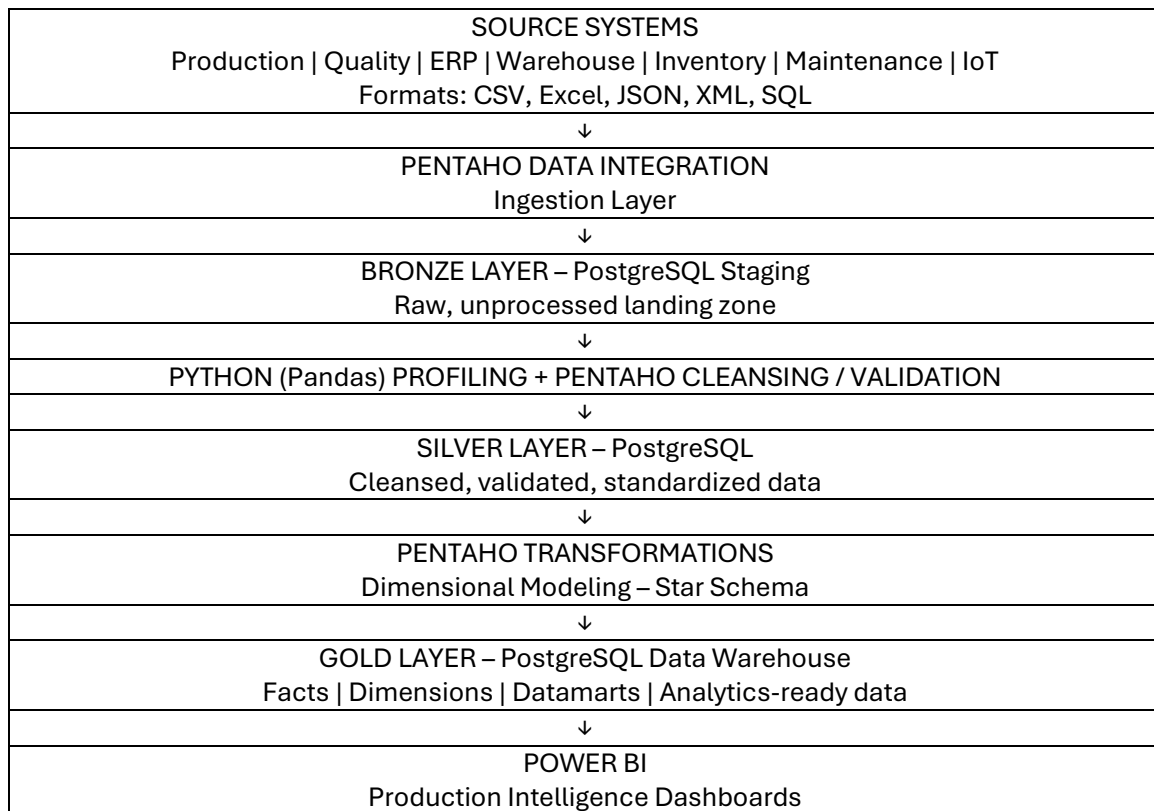
SOLUTION ARCHITECTURE

Enterprise Manufacturing Data Platform (EMDP)

1. Architecture Overview

The Enterprise Manufacturing Data Platform (EMDP) follows the Medallion Architecture pattern, progressively refining data through three layers: Bronze (raw), Silver (cleansed), and Gold (curated and business-ready). Pentaho Data Integration orchestrates data movement between layers, PostgreSQL serves as the storage engine, Python performs data profiling, Power BI provides the final analytics experience, and Git/GitHub provides version control for pipeline code and documentation.

2. High-Level Architecture



3. Layer Descriptions

Layer	Purpose	Storage
Bronze	Raw ingested data with no transformation applied; audit columns such as load_timestamp, source_file,	PostgreSQL – bronze schema

	and batch_id are added.	
Silver	Cleansed, validated, standardized, and deduplicated data.	PostgreSQL – silver schema
Gold	Star schema with fact and dimension tables, curated datamarts, and analytics-ready data.	PostgreSQL – gold schema

4. Technology Stack

Category	Technology
ETL	Pentaho Data Integration (Spoon)
Database	PostgreSQL
Programming	Python (Pandas)
Version Control	Git & GitHub
Reporting	Power BI
Documentation	MS Word
Project Management	Agile Scrum

5. Repository Structure

The repository is organized according to the Medallion Architecture and the project specification:

enterprise-manufacturing-data-platform/

└─ docs/ – BRD, project charter, architecture, and project documentation

└─ config/ – database connections and environment configuration

└─ datasets/ – sample and raw source files

└─ bronze/

| └─ ingestion/

| └─ staging/

└─ silver/

| └─ cleansing/

| └─ validation/

| └─ transformations/

└─ gold/

```
|  ├── warehouse/
|  ├── datamarts/
|  └── analytics/
    ├── pentaho/
    ├── transformations/
    └── jobs/
python/ – profiling and cleansing scripts
sql/ – DDL and DML scripts
metadata/ – data dictionary and business glossary
lineage/ – source-to-target mappings
dashboards/ – Power BI files
deployment/
tests/
└── README.md
```

6. Data Flow Summary

- Source systems generate data in CSV, Excel, JSON, XML, and SQL formats.
- Pentaho transformations ingest each source into PostgreSQL Bronze staging tables.
- Python scripts profile Bronze data to identify quality issues.
- Pentaho cleansing transformations move validated data into the Silver layer.
- Pentaho jobs populate Gold-layer Fact and Dimension tables using a Star Schema.
- Power BI connects to the Gold layer to deliver production intelligence dashboards.
- Metadata, lineage, and Git commit history are maintained throughout the flow.

7. Governance Considerations

- Every table is documented in the Data Dictionary.
- Source-to-target mappings are recorded for full lineage traceability.
- All pipeline code is committed to Git with meaningful, sprint-aligned commit messages.

8. Architecture Principles

- Layered data processing using Bronze, Silver, and Gold.

- Centralized storage using PostgreSQL.
- Reusable and orchestrated ETL pipelines using Pentaho.
- Data profiling and quality assessment using Python.
- Traceability through metadata, lineage, audit fields, and Git version control.
- Separation of raw, cleansed, and business-ready data.